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Towards Robust and Explainable Automatic Depression Detection with Foundation Models

Graduation Project Proposal (Artificial Intelligence)

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1 Introduction

Depression is a leading contributor to global disability and economic burden (Organization, 2017), and phenomenologically characterized by profound existential suffering and social disconnection (Ratcliffe, 2014). Because untreated depression reduces the probability of remission (Ghio et al., 2014; Kraus et al., 2019), timely identification is critical. The gold-standard (Smithson & Pignone, 2017) clinical interview is resource-intensive and frequently under-utilized in primary care (First et al., 2015; Brodey et al., 2018). Since depression seems to affect the neural processes underlying communication (Aloshban et al., 2022), automated speech analysis using machine learning has emerged as a promising, non-invasive diagnostic tool (Cummins et al., 2015; Wu et al., 2023; Kim et al., 2023; Donaghy et al., 2024; Briganti & Lechien, 2025; Liu et al., 2024). In this project, we seek to boost progress towards the development of explainable systems that can differentiate between depressed and non-depressed individuals.

2 Theoretical Framework

A common paradigm in automated depression detection is the use of classical machine learning techniques on hand-crafted feature sets (Kim et al., 2023). Examples of speech features that are found to be associated to depression are: fundamental frequency (F_0), energy, speech rate, pause measures, jitter, shimmer, Harmonic-to-Noise Ratio (HNR), Normalised Amplitude Quotient (NAQ), Quasi-Open-Quotient (QOQ), Formant Frequencies, Mel-Frequency Cepstral Coefficients (MFCCs), Spectral Tilt (Cummins et al., 2015), Linear Predictive Cepstrum Coefficients (LPCCs) (Wu et al., 2023), those from eGeMAPS (Extended Geneva Minimalistic Acoustic Parameter Set), and those from COMPARE (Interspeech Computational Paralinguistics Challenge) (Kim et al., 2023). Though the use of hand-crafted features is valuable with respect to explainability, one limitation of this method is that useful information related to depression might be lost when using these feature sets (Kim et al., 2023; Ma et al., 2016). Considering this, researchers have started developing deep learning based ADD (Leal, Ntalampiras, & Sassi, 2025). These methods are notoriously data-intensive. Due to the limited availability of publicly available datasets for ADD (Leal, Ntalampiras, & Sassi, 2025), the use of foundation models as feature extractors is particularly useful in this regard, as these models are trained through Self-Supervised Learning (SSL), i.e., by solving a task where the supervisory signal is derived from the input data itself (Brima et al., 2024), partly bypassing the need for massive amounts of labeled data. In the context of speech processing, a foundation model may use thousands of hours of unlabeled (transcriptions of) speech data to learn underlying relevant features of speech. The representations that these models use could then be used for a variety of tasks, including automatic depression detection.

Two examples of SSL models for speech signals are Wav2Vec 2.0 and HuBERT (trained on LibriSpeech and Libri-Light (Hsu et al., 2021)). Both models contain, among other elements, a convolutional feature encoder, followed by a transformer encoder (Baevski et al., 2020; Hsu et al., 2021).



The convolutional encoder transforms the raw waveform into a latent representation, and the transformer encodes context representations that capture long-range dependencies. As a result of training on large corpora, these resulting embeddings are rich in information about a wide range of features of speech (including phonology, prosody, intonation, etc) (Baevski et al., 2020; Hsu et al., 2021). Still, one problem remains with the use of these models, which is that of domain mismatch, often referred to as covariate shift (Shimodaira, 2000). The pre-trained model, having learned its representation from a broad source domain such as publicly available speech data, may produce features whose distribution ($\mathbf{P}(\mathbf{X})$) is suboptimal or misaligned when applied to the speech data of an ADD task. If the model's latent features lack the required sensitivity to capture subtle cues of psychological phenomena, there is a distributional mismatch in the features it extracts. A solution to this problem would be to perform fine-tuning: by adapting the model's parameters with a small amount of target-specific data, the model can adjust its encoder to optimize its representations for the current domain (Pan & Yang, 2010).

Several studies have used foundation models in some way to examine psychopathological phenomena. For example, Gao et al. (2025) use embeddings from speech foundation models fed into a two-layer MLP classifier for suicide risk detection, consistently obtaining F1 scores over 0.6. Brueckner et al. (2025) used Support Vector Machines, Random Forests, eXtreme Gradient Boosting decision tree algorithm and fully connected deep neural networks on Wav2Vec 2.0 embeddings to detect generalized anxiety disorder and major depressive disorder, with performance above - though not by much - chance level in 93,75% of configurations. Zhang et al. (2024) combined Wav2Vec 2.0 with an additional structure (a one-dimensional CNN, attention pooling layer, an LSTM, and self-attention mechanism) and simultaneously fine-tuned Wav2Vec 2.0's context encoder and trained the downstream layers on a subset of the DAIC-WOZ dataset (Gratch et al., 2014) and the CMDC dataset (Zou et al., 2023) and achieved an F1 score of 0.79 on the test set from the DAIC-WOZ and 0.91 on that of the CMDC for depression detection. Huang et al. (2024) used a fine-tuning network (consisting of a maxpool, a fully connected layer and two dropout layers) downstream from a frozen Wav2Vec 2.0 encoder for depression detection, and achieved a 0.19 square root error rate for binary classification and 0.38 "RMSE" (square root of average squared category distance/error per sample) for multiclass (four levels) classification. Finally, perhaps most related to this project would be the work of Leal, Ntalampiras, Rossetti, et al. (2025), who extracted MFCCs and Wav2Vec 2.0 embeddings from recordings of cognitive behavioral therapy sessions, and fed them into five different low-parameter models (Random Forest [RF], XGBoost, Support Vector Regression [SVR], Logistic Regression [LR], and a MultiLayer Perceptron [MLP]) to detect depression severity (PHQ-9 score) - they obtained a recall of 70% for detecting improvement versus worsening in young adults.

Though some studies do fine-tune layers upstream from their classification head (in lieu of merely using the foundation models as a frozen feature extractor), and some studies do employ models trained on publicly available datasets in new contexts that the model has not seen before, to our knowledge, no study has yet combined these two approaches. Further, little attention has been paid to explaining the decision making of these models. Hence, in this study, we will train a foundation model on a publicly available dataset, use it to predict depression severity in a psychological intervention study, and then shed light on how this model makes decisions by relating the model's



decision making to acoustic features that were previously found to potentially mark depression severity.

3 Research Question

The research question (RQ) that runs through this project is: *how well does a foundation-model-based depression detection system fine-tuned end-to-end on a public dataset generalize to a novel intervention context, and which established acoustic features might explain its decision making?* Answering this question involves four phases for the project:

- **Phase 1: State of the art.** In this part of the project, we briefly review the state of the art in automatic depression detection. Here, we identify gaps in the literature. The **sub-RQ** is: *What work has been done in detecting depression (or other psychopathology) in speech using foundation models, and what work remains to be done?*
- **Phase 2: Model Design and Training.** In this part, we choose a deep learning architecture that would best fit the findings from Phase 1, we implement and train this model on the WOZ-DAIC dataset (Gratch et al., 2014). The **sub-RQ** is: *Given data characteristics, which model and training strategy would have the best chance of generalizing to a novel context?*
- **Phase 3: Model Generalization.** Here, we evaluate the performance of this model on entirely new data obtained from a mindfulness-based intervention study. We interpret and discuss the results. The **sub-RQ** is: *How well does the system generalize to a novel intervention context?*
- **Phase 4: Model Decision making.** Finally, we seek to decrease the opacity of the model's decision making by examining the model's behavior in relation to previous findings in automatic depression detection literature. The **sub-RQ** is: *What information does the model use in its decisions, and how does it relate to previous findings?*

4 Methods

To develop and evaluate the depression detection system, this project follows a four-phase methodology aligned with the research questions.

- **Phase 1:** To examine where existing methodologies fall short, we conduct a critical review of ADD studies to identify ways in which these could be improved. Here, we include all studies that feature the detection of depression in speech using foundation models.
- **Phase 2:** We develop the depression detection system by appending a classification/regression (depending on how 'depression' is operationalized) head (selected through a limited grid search over hyperparameters) to a pre-trained transformer-based speech model. Rather than



using the model as a frozen feature extractor, we conduct end-to-end fine-tuning on the WOZ-DAIC dataset, allowing the model’s transformer layers to adapt its context representations to the specific distributional characteristics of depressive speech. To ensure the model adapts to the nuances of depressive speech without losing its general understanding of speech, we use a layered learning rate: the classification head is initialized with a standard rate (e.g., 10^{-3}), while the transformer backbone utilizes a significantly lower rate (e.g., 10^{-5}), and both are adapted through a cosine annealing scheduler.

- **Phase 3:** To evaluate the system’s sensitivity to symptom changes, we test the model on longitudinal data from a two-week mindfulness-based intervention study ($N \approx 50$), as such interventions are known to reduce depressive symptoms (Alkan et al., 2025). Participants provide high-quality spontaneous speech recordings in a controlled lab setting (pre- and post-intervention) alongside daily smartphone-based voice reflections to ensure ecological validity (Shin & Bae, 2024). This diverse longitudinal dataset allows us to validate the system’s ability to track depression-related features over time and evaluate to what extent changes in the system’s predictions correlate with changes in actual symptoms.
- **Phase 4:** To examine the model’s behavior in relation to previous findings in automatic depression literature, we will generate three binary masks over all samples that mark important information regarding three previously established features: F_0 variation (calculated over a small sliding window), MFCC 2 (extreme values, e.g. $> 1\sigma$), and pauses in speech. We then generate local salience maps over the time-frequency domain (Kechris et al., 2025) for the depression classification decisions to obtain the top 10% salience bins for every sample. We then calculate the degree in overlap between this mask and the union over the three previously generated binary masks.

5 Scientific Relevance for Artificial Intelligence

By testing the potential of foundation-model-based models to detect depression in the context of a psychological intervention, this project contributes to methodological innovation in computational psychiatry. This project tests the empirical limits of cross-domain generalization by assessing how well representations fine-tuned on public datasets transfer to a longitudinal, real-world intervention setting. Rather than opening a completely new research line, this project is a natural extension of extant research, combining several approaches from similar studies. The most innovative aspect of our project is the comparison between the model’s predictions and acoustic features that have previously been found to be related to depression; in doing so, this project contributes to the development of *explainable* decision support tools within the field of artificial intelligence applied to mental health.



6 Planning

The planning of the graduation project is roughly as follows:

- March 10, 2025 – August 25, 2025: **Project Initiation**. Conceptual development, development of psychological intervention study protocol, establishing preliminary theoretical background, drafting machine learning pipeline for detecting acoustic differences between intervention groups.
- July 8, 2025 – February 1, 2026: **Bachelor's Project (Phase 1)** Collaboration in development and clinical background of machine learning project for automatic depression detection, compile and critically assess candidate datasets, help interpret findings from data exploration and final pipeline.
- August 25, 2025 – December 31, 2025: **Pilot Study (Phase 3)**. Evaluating collection/processing/storage protocol for intervention study data, designing mindfulness-based intervention and control condition, producing daily self-guided audio files, revise based on feedback, recruiting participants, revising study protocol and iterative testing.
- January 1, 2026 – March 1, 2026: **Final Scoping (Phase 1 & Phase 2)**. Theoretical framework of project is revised and then refined to align more closely with empirical precedent, findings from initial implementation and exploration of experimental data are used to finalize conceptual outline and project planning.
- January 1, 2026 – April 30, 2026: **Collect Data (Phase 3)**. Carrying out mindfulness-based intervention and collecting data.
- March 1, 2026 – April 30, 2026: **Implementation (Phase 2)**. Implement preprocessing, feature extraction, fine-tuning according to training schedule, evaluation metrics for test data, global and local explainability. Revise based on technical feasibility.
- March 11, 2026 – May 5, 2026: **Writing (Phase 1 & Phase 2)**. Writing first draft of methods, to be handed in before April 7th, and that of the introduction section, handed in before May 5th.
- May 6, 2026 – May 18, 2026: **Model Generalization & Explainability (Phase 3 & 4)**. Test final model on and calculate results for data from intervention study. Store all relevant plots and statistics, delete all other data. Submit draft for results section before May 19th.
- May 15, 2026 – June 1, 2026: **Critical Review**. Critically evaluate findings from analysis of the model's decision making, submit draft of discussion section before June 2nd.
- June 1, 2026 – July 5, 2026: **Finalize**. Finalize thesis, present findings, submit final version

7 Resources and Support

The graduation project will be carried out partly on campus of the Faculty of Science and Engineering of the University of Groningen, but mostly from home. On campus, the author of this proposal will carry out the explanation of the intervention to participants, and the real-time administration of the intervention. A staff microphone will be used to record interviews with participant, and participants record post-practice reflections on their personal smartphone. For fine-tuning the deep learning



model on the DAIC-WOZ, I request access to the GPU servers of the Artificial Intelligence department of the University of Groningen. For automatic transcription of voice data, I request access to the Hábrók HPC Cluster Web Portal. This access ensures compliance, as all sensitive data processing will remain on the university's secure environment. Supervision will consist of a weekly supervision meeting of 30 minutes, up to an hour if necessary.

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